Coordinate Geometry: Linear Graphs

Questions 1–24

Question 1. Find the line passing through each of the following pairs of points giving your answer in the form $ax + by = c$ where a, b and c are integers:

a) $(2, 5)$ and $(1, 7)$
b) $(1, 4)$ and $(4, 13)$
c) $(-1, -3)$ and $(5, 1)$
d) $(-4, -6)$ and $(-2, -17)$
e) (p, q) and $(2p, 4q)$
f) $(p, \frac{1}{p})$ and $(\frac{1}{p}, p)$ where $p \neq -1, 0, 1$

Question 2. State both the gradient of the line and the gradient of a line which is perpendicular to each of the following:

a) $3x + 6y = 7$
b) $3y - 2x = 1$
c) $2x + 8y = 11$
d) $7y - 5x - 2 = 0$
e) $20x + 17y = 3$
f) $ax + by = c$

Question 3. For each of the following pairs of lines, determine if they are perpendicular to each other:

a) $2x + 3y = 1$ and $y = -\frac{3}{2}x + 5$
b) $x - 5y + 10 = 0$ and $y = 5x - 4$
c) $24x + 12y = 16$ and $12y + 24x = 1$
d) $y = -4x + 8$ and $4y - x = 28$
e) $ax + by = p$ and $bx - ay = q$

Question 4. For the lines $ax + by = c$ and $px + qy = r$ where a, b, c, p, q and r are integers find a relationship between a, b, p and q such that the two lines are

a) Parallel.
b) Perpendicular.

Question 5. Find the midpoint between each of these pairs of points:

a) $(-4, 3)$ and $(4, 9)$
b) $(1, 7)$ and $(6, -5)$
c) $(-8, 10)$ and $(-2, -1)$
d) $(-7, -2)$ and $(\frac{7}{2}, 12)$
e) (p, q) and $(2p, 3q)$
f) $(p, \frac{1}{p})$ and $(\frac{1}{p}, p)$ where $p \neq -1, 0, 1$

Question 6. For each of the following, find the value of k given the point which lies on the line:

a) $(k, 4)$ and $2y - 5x = 16$
b) $(2k, k)$ and $4x - 3y + 20 = 0$

Question 7. Find the perpendicular bisector of the each of the following points giving your answer in the form $ax + by = c$ where a, b and c are integers:

a) $(-2, 5)$ and $(6, 1)$
b) $(3, 5)$ and $(6, 8)$
c) $(1, 2)$ and $(4, 10)$
d) $(-6, 2)$ and $(-1, -5)$
e) $(-9, 11)$ and $(-1, -2)$
f) (p, q) and $(2p, 3q)$

Question 8. Find the equation of the straight line which is parallel to the line $2y - x = 5$ and passes through the point $(-1, 4)$. Give your answer in the form $ax + by = c$ where a, b and c are integers.

Question 9. Find the equation of the straight line which is perpendicular to the line $6y + 2x - 11 = 0$ and passes through the point $(2, 3)$. Give your answer in the form $ax + by = c$ where a, b and c are integers.

Question 10. Find the angle (in degrees to 1 decimal place) between each of the following pairs of lines:

a) $y = 0$ and $y = 2x$
b) $y = 3x + 3$ and $y = 4x - 1$
c) $y = 5x + 8$ and $y = -x$
d) $y = -2x - 4$ and $y = -5x + 1$

Question 11. Find the area of the triangle bounded by the coordinate axes and the line $2y + x - 11 = 0$.

Question 12. Find the area in terms of k of the triangle bounded by the coordinate axes and the line $ky + 3x = 2$ where k is a positive constant.

Question 13. The three points $(3, 11)$, $(4, -2)$ and $(-2, 1)$ are all corners of a triangle. By considering the gradients of the lines passing through each of the pairs of points, show that it is a right-angled triangle.

Question 14. The three points $(5, 10)$, $(-1, 8)$ and $(4, -7)$ are all corners of a triangle. By considering the distances between each of the pairs of points, show that it is a right-angled triangle and hence find the area of the triangle.

Question 15. Find the value of k such that $2y - 7x = 0$ and $ky + 6x = 10$ are perpendicular to each other.

Question 16. A line perpendicular to $y + 2x = 9$ passes through $(p, 6)$ and $(1, 7)$. Find the value of p.

Question 17. Two points $A(p, -2)$ and $B(5, q)$ have a perpendicular bisector with equation $3y + x = 15$. Find the values of p and q.

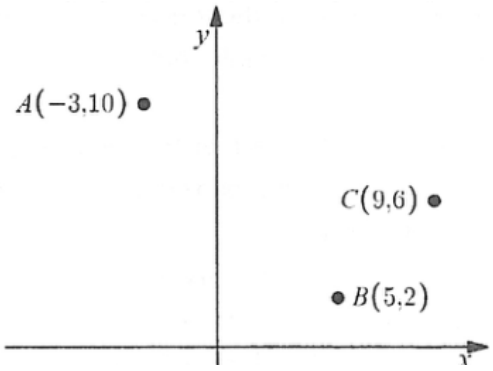
Question 18. Show that $(-5, 7)$, $(10, 10)$ and $(-15, 5)$ are collinear.

Question 19. Find the value(s) of k such that

a) $(-15, 9)$, $(3, 3)$ and $(k, -5)$ are collinear.

b) $(1, k)$, $(-3k, 3)$ and $(7, \frac{3}{2})$ are collinear.

Question 20. Three points A, B and C have coordinates $(-3, 10)$, $(5, 2)$ and $(9, 6)$ respectively.



a) Find the equation of the perpendicular bisector of A and B in the form $y = mx + c$.

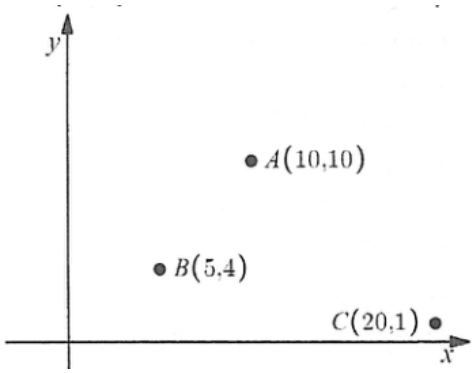
b) Find the equation of the perpendicular bisector of B and C in the form $y = mx + c$.

c) Find the point P, where the two perpendicular bisectors intersect.

d) Find the distance between the point P and each of the three points.

Question 21. By considering the line that is perpendicular to $3y - 4x = 5$ which passes through the point $P(11, 8)$, find the shortest distance between $P(11, 8)$ and the line $3y - 4x = 5$.

Question 22. Three points $A(10, 10)$, $B(5, 4)$ and $C(20, 1)$ and $D(15, k)$ form the vertices of a trapezium.



a) Show that AB is not perpendicular to BC.

Find

b) the equation of the line passing through B and C in the form $y = mx + c$

c) the equation of the line passing through A and D in the form $y = mx + c$ given that it is parallel to the line passing through points B and C

d) the value of k

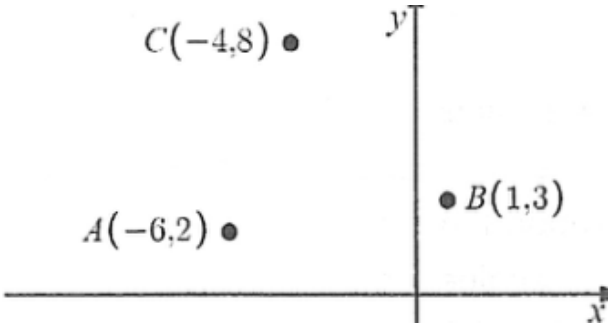
e) the distances between A & D, and B & C

f) the equation of the line which is perpendicular to BC and passes through A

g) the shortest distance from A to the line BC

h) the area of the trapezium $ABCD$

Question 23. The points $A(-6,2)$, $B(1,3)$, $C(-4,8)$ and D form a quadrilateral.



a) Show that the distances between A and B, and between B and C are the same.
The line AC is perpendicular to the line BD.
b) Show that the possible coordinates of the point D satisfy the line $x + 3y = 10$.

Question 24. Four points $A(-3,-1)$, $B(2,-\frac{3}{5})$, $C(4,4)$ and $D(-1,\frac{18}{5})$ form a quadrilateral. By considering the diagonals only, AC and BD, determine its type.